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Identification of Dysfunctional Gut Microbiota Through Rectal Swab in Patients with Different Severity of Acute Pancreatitis

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Abstract

Background Acute pancreatitis (AP) has a wide spectrum of severity and can be associated with considerable morbidity and mortality. Whether gut microbiota dysbiosis is associated with AP severity remains obscure.

Aims We aim to investigate the differences in the alterations of gut microbiota in different grades of AP severity.

Methods We collected clinical information and rectal swab samples from 80 individuals. The gut microbiota was tested by 16S rRNA gene sequencing, gut microbiota species composition analysis, difference analysis, random forest model prediction analysis, and gut microbiota species correlation network analysis.

Results There was a different microbiota profile in different severity grades. *Bacteroides*, *Escherichia-Shigella*, and *Enterococcus* were dominant species in mild, moderately severe, and severe AP, respectively. *Finegoldia* was the most significantly increased and *Blautia* the most decreased species in mild AP. *Anaerococcus* was the most significantly increased and *Eubacterium hallii* the most decreased species in moderately severe AP. *Enterococcus* was the most significantly increased and *Eubacterium hallii* the most decreased species in severe AP. *Finegoldia*, *Eubacterium hallii*, and *Lachnospiraceae* were potential diagnostic biomarkers for mild AP and *Eubacterium hallii* and *Anaerococcus* for moderately severe AP. There was a positive interaction between *Firmicutes* and *Bacteroidetes* in mild AP.

Conclusions The disturbed gut microbiota is different among grades of AP, suggesting their potential role in the progression of disease severity.

Graphic Abstract

There was a different microbiota profile in different severity grades. *Bacteroides*, *Escherichia-Shigella*, and *Enterococcus* were dominant gut microbiota species in MAP, MSAP, and SAP, respectively. *Finegoldia* was the most significantly increased

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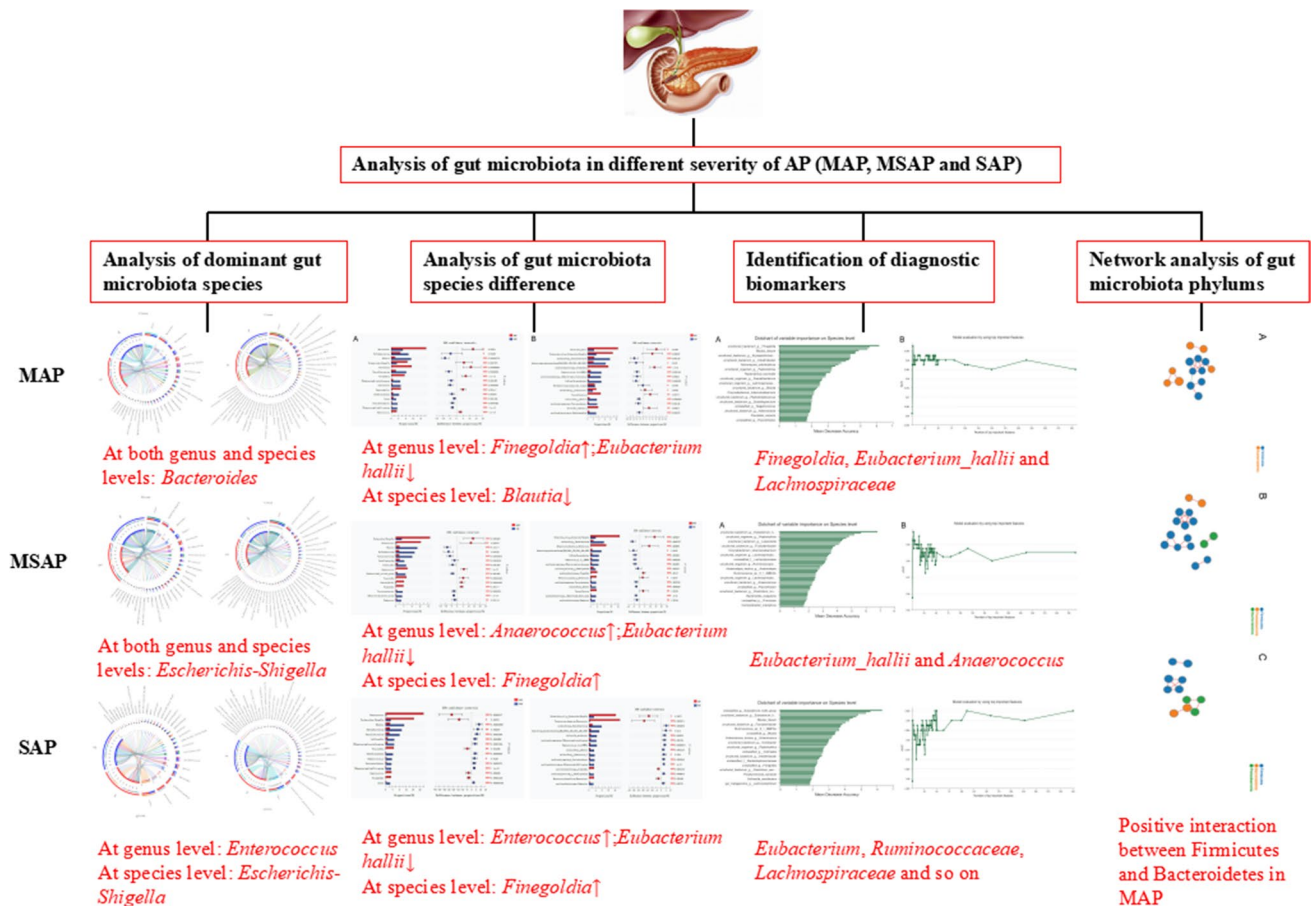
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and *Blautia* the most decreased gut microbiota species in MAP. *Anaerococcus* was the most significantly increased and *Eubacterium hallii* the most decreased species in MSAP. *Enterococcus* was the most significantly increased and *Eubacterium hallii* the most decreased species in SAP. *Finegoldia*, *Eubacterium hallii*, and *Lachnospiraceae* were potential diagnostic biomarkers for MAP and *Eubacterium hallii* and *Anaerococcus* for MSAP. There was a positive interaction between *Firmicutes* and *Bacteroidetes* in MAP.



Keywords Acute pancreatitis · Severity · Gut microbiota · Microbiota dysbiosis · Diagnostic biomarkers

Introduction

As the most common digestive disorders leading to acute admission, acute pancreatitis (AP) carries a significant clinical and economic burden, and there is a broad spectrum of severity [1]. The most commonly used method for grading the severity of AP is the revised Atlanta Classification, which classifies AP into mild (MAP), moderately severe (MSAP), and severe (SAP) [2]. Drivers of AP severity include the pancreatic necrosis, the inflammatory response, and organ dysfunction. Patients with SAP have an increased systemic inflammatory response and a higher incidence of transient and persistent organ failure, with the latter being the leading cause of death [3–6]. The role of the intestine in the pathogenesis of AP and other acute and critical illness is

essential but often overlooked. The loss of gut barrier function leads to bacterial and endotoxin translocation, which in turn plays a vital role in generating the second inflammatory hit [7]. For example, Johnson et al. [8] found that gut bacteria translocation (via hematogenic, lymphatic, and reflux from the duodenum and the biliary tree) was involved in infections in AP. It indicates a probable association between gut microbiota and AP progression.

The term “microbiota” refers to the microbial flora harbored by humans, including symbiotic, commensal, and pathogenic microorganisms. Alterations in human intestinal microbiota homeostasis (dysbiosis) are associated with the systematic inflammatory response syndrome (SIRS) and a range of diseases [9]. Researchers have found different gut microbiota in AP patients with different severity

gradings [10]. They uncovered a significant increase in *Acinetobacter*, *Stenotrophomonas*, and *Geobacillus* and a substantial reduction in *Bacteroides*, *Alloprevotella*, *Blautia*, and *Gemella* in SAP than MAP and MSAP. This mechanism gives credence to the potential association between intestinal dysbiosis and AP severity. It is presumed that after the onset of AP, intestinal permeability is increased due to mucosal ischemia and the disruption of the gut barrier [11, 12]. Then gut bacteria could cross the impaired gut barrier to reach the pancreas and systemic circulation, leading to infected pancreatic necrosis and contributing to mortality risk in AP [11–13]. What remains obscure is whether different dysbiosis occurs with AP progression, especially in the process of MAP, MSAP, and SAP separately.

The 16S rRNA gene, presented in the genome of all bacteria, is highly conserved and specific. As a result, 16S rRNA gene sequencing is used extensively to study gut microbiota in a number of diseases [14, 15]. In this study, we used 16S rRNA gene sequencing followed by a series of other analyses to investigate the intestinal microbiome in patients with AP and to determine whether any changes are related to the different grades of severity.

Methods

Patient Populations

The 2012 revised Atlanta classification stratified the clinical severity of AP patients into three categories: mild AP (MAP), moderately severe AP (MSAP), and severe AP (SAP) [2]. From June 2018 to April 2019, all the patients with a diagnosis of AP based on the 2012 Revised Atlanta Classification and admitted to Peking Union Medical College Hospital, Beijing, China, were eligible for inclusion, if they were enrolled within 72 h of the onset of symptoms. A total of 20 MAP, 20 MSAP, and 20 SAP patients were ultimately included.

Normal Control Population (NOR)

These NOR were healthy volunteers (community workers and students) identified from Peking Union Medical College Hospital. They were age and gender matched to the patient population and selected after the completion of the patient study. The NOR group had to fulfill the following inclusion criteria: free of metabolic, cardio-cerebrovascular, and intestinal diseases; no pregnancy; no known allergens; no medical treatment influencing bowel function; not eating a diet specific to weight loss or other particular purposes [10].

Exclusion Criteria

Patients and NOR were excluded if they took antibiotics, probiotics, Chinese herbs, steroids, and other substances that may influence the structure of the flora in up to 8 weeks prior to fecal sampling. Other exclusion criteria referred to patients with a history of immunodeficiency, allergy, asthma, celiac disease, colon cancer, diabetes, HIV, inflammatory bowel disease, irritable bowel syndrome, gastroenteritis, necrotizing enterocolitis, and arthritis [10]. In addition, none of all AP patients had been addressed in other hospitals.

Written informed consent was obtained from each participant. This study was approved by the institutional review boards of Peking Union Medical College Hospital (No. JS 1826).

Analysis of Clinical Characteristics of AP Patients and Healthy Subjects

In view of the multiple etiology of AP, we first performed the corrected analysis based on several etiology factors to search for differences at baseline in this study. Then, the systemic inflammatory response syndrome (SIRS) was assessed as follows: The diagnosis of SIRS must meet at least two of the following four aspects: (1) temperature $> 38^{\circ}\text{C}$ or $< 36^{\circ}\text{C}$; (2) heart rate > 90 beats/min; (3) respiratory rate > 20 breaths/min or $\text{PaCO}_2 < 32$ mmHg; (4) white blood cell count $> 12 \times 10^9/\text{L}$ or $< 4 \times 10^9/\text{L}$ ($> 12,000/\text{ul}$ or $< 4000/\text{ul}$) [16]. In addition, the infectious pancreatic necrosis and sepsis were further assessed. According to the 2012 Atlanta classification, the diagnosis of infectious pancreatic necrosis is primarily by enhanced CT and fine needle aspiration (FNA) [2]. Bubbles are present in the extra luminal pancreatic or peripancreatic tissue in the enhanced CT test. Bacteria and/or fungi are present in the FNA analysis. The diagnosis of sepsis is made when infection is present and $\text{SOFA} \geq 2$ [17].

Sample Collection, DNA Extraction, PCR Amplification, and Illumina MiSeq Sequencing

Stool samples are generally considered the reference standard but are not practical to obtain in patients with AP, who are usually complicated with impaired gastrointestinal motility [17], and thus, rectal swabs are often used instead [18]. In this study, microbial DNA was extracted from fecal samples obtained through the rectal swab immediately after admission. All individuals should have an empty stomach (did not eat for 12 h) before sample collection. The sample of normal healthy volunteers was collected in the morning. The timing of the rectal swab sampling of AP patients was within 24 h after disease onset in all individuals. Detailed processes of fecal sample collection were as follows: (1) Soap, water,

and 70% alcohol were used to clean around the anus. The disinfectant was evaporated on its own to reduce skin commensals contamination; (2) the sterile swab was infiltrated with normal saline for 2 min before inserting into the anus as deep as 4–5 cm (referred to previous report [19]) and was then rotated gently at the anal sphincter to get the fecal samples; (3) the swab with fecal samples was quickly placed into the sterile tube and stored at the -80°C refrigerator for RNA extraction, PCR amplification, and Illumina MiSeq sequencing.

Sample Comparative Analysis in MAP, MSAP, and SAP

In the sample comparative analysis, beta diversity analysis (sample-level cluster analysis) and sample grouping analysis (partial least squares discriminant analysis, PLS-DA) were included. In the beta diversity analysis, Bray–Curtis distance algorithm and the average distance method were used for hierarchical clustering analysis. In the sample grouping analysis, PLS-DA can effectively distinguish the observed values between groups and find the influencing variables that lead to the differences between groups.

Annotation and Evaluation of Gut Microbiota Species in MAP, MSAP, and SAP

In order to understand the abundance and uniformity of gut microbiota species, changes in total and core gut microbiota species and diversity of community, we, respectively, performed rank-abundance, pan/core, and alpha diversity analyses by using R language.

Analysis of Gut Microbiota Species Composition in MAP, MSAP and SAP

In MAP, MSAP, and SAP, Venn diagram, Heatmap, and Circos diagram were, respectively, used to analyze the gut microbiota species composition, community composition, and the relationship between samples and gut microbiota species by using R language.

Analysis of Gut Microbiota Species Difference in MAP, MSAP, and SAP

Based on the community abundance data, relevant analysis methods were used to detect the diversity of microbial communities. In this study, the significance test was used to screen the different gut microbiota species in MAP, MSAP, and SAP.

Model Prediction Analysis in MAP, MSAP, and SAP

In the model prediction analysis, the random forest model was utilized to select the most important diagnostic biomarkers efficiently and rapidly through random forest package in R language. The detailed process was as follows: Firstly, the importance of value of each differential gut microbiota species in accuracy is ranked according to the mean decrease. Then, the optimal number of features was found by subsequently adding one differential gut microbiota species at a time in a top-down forward-wrapper approach.

Network Analysis in MAP, MSAP, and SAP

In order to assess whether there are interactions between gut microbiota species that belonged to the same phylum in MAP, MSAP, and SAP. The gut microbiota species correlation network analysis at the phylum level was performed. Herein, top 50 gut microbiota species in total abundance of classification level were selected. The correlation was calculated by the Spearman method, and the correlation coefficient $|r| > 0.8$ was set to establish the correlation network.

Results

Clinical Characteristics of AP Patients and Healthy Subjects

A total of 60 AP patients (MAP 20, MSAP 20, and SAP 20) and 20 NOR were enrolled in this study as the etiology for AP (such as alcohol consumption) makes a huge difference to the microbiota involved. For example, Philips et al. [20] reported the differences between gut microbiota among alcohol users who developed acute alcoholic hepatitis compared to alcoholic pancreatitis. Hyperlipidemia is another important etiologic factor for AP [21]. Therefore, we performed the corrected analysis based on these etiologic factors to search for differences at baseline in this study. The clinical characteristics of AP and NOR group after corrected analysis are presented in Table 1. There were no significant differences between MAP, MSAP, and SAP in terms of etiologic factors (biliary, alcohol consumption, and hyperlipidemia). Moreover, all these individuals did not have hypothyroidism. The level of inflammation marker (CRP) in SAP was significantly greater than that in MAP and MSAP. SIRS was absent in MAP patients but was present in patients with MSAP and SAP (Table 1).

Sample Comparative Analysis in MAP, MSAP, and SAP

From beta diversity analysis (sample-level cluster analysis), we found the correlation between samples of MAP

Table 1 General characteristics of patients

Variables	NOR	MAP	MSAP	SAP	Z/χ^2	<i>p</i>
Number of patients	20	20	20	20		
Age, years	32(29,46)	47 (38,58)	43 (34,60)	44 (31,50)	$\chi^2=7.186$	0.091
Gender/male, <i>n</i> (%)	11 (55.0)	10 (50.0)	8 (40.0)	13 (65.0)	$\chi^2=2.607$	0.456
BMI	22.9 (20.3,24.8)	25.7 (24.1,27.8) ^a	27.5 (24.4,30.1) ^a	26.0 (24.1,28.0) ^a	$\chi^2=18.22$	<0.001
APACHE II score		2 (0,4.5)	3 (2.3,6)	9 (8,11) ^{bc}	$\chi^2=35.78$	<0.001
Balthazar CT score ($\geq D$), <i>n</i> (%)		0 (0.0)	8 (40.0) ^b	19 (95.0) ^{bc}	–	<0.001
Smoking <i>n</i> (%)		4 (20.0)	2 (10.0)	4 (20.0)	–	0.750
Drinking history, <i>n</i> (%)	4 (20.0)	5 (25.0)	5 (25.0)	5 (25.0)	–	1.000
CRP		106.9 (11.5151.4)	156.9 (124.0160.0) ^b	160.0 (160.0256.0) ^{bc}	$\chi^2=21.89$	<0.001
Etiology, <i>n</i> (%)					–	0.731
Biliary		8 (40.0)	5 (25.0)	8 (40.0)		
Hyperlipidemia		7 (35.0)	10 (50.0)	9 (45.0)		
Alcohol consumption		5 (25.0)	5 (25.0)	3 (15.0)		
Outcomes						
Organ failure, <i>n</i> (%)		0 (0.0)	12 (60.0) ^b	20 (100.0) ^{bc}	$\chi^2=40.71$	<0.001
Organ failure duration (h)			32 (14,40)	58 (50,74)	$Z=2.81$	0.005
Systematic complications, <i>n</i> (%)			12 (60.0)	20 (100.0)	–	0.003
ACS			1 (5.0)	2 (10.0)	–	1.000
SIRS			3 (15.0)	19 (95.0)	$\chi^2=25.859$	<0.001
ARDS			8 (40.0)	14 (70.0)	$\chi^2=3.636$	0.057
AKI			3 (15.0)	8 (40.0)	$\chi^2=3.135$	0.077
Shock			1 (5.0)	7 (35.0)	–	0.044
Liver damage			1 (5.0)	4 (20.0)	–	0.342
Myocardial injury			0 (0.0)	2 (10.0)	–	0.487
Disorder of consciousness			0 (0.0)	1 (5.0)	–	1.000
Sepsis		0 (0.0)	3 (15.0)	13 (65.0)		<0.001
ICU, <i>n</i> (%)		0 (0.0)	4 (20.0)	19 (95.0) ^{bc}	$\chi^2=42.44$	<0.001
Hospital stay (days)		5 (3,6)	12 (7,18) ^b	27 (18,34) ^{bc}	$\chi^2=35.78$	<0.001
ICU stay (days)			0 (0,0)	7.5 (5.0,12.5)	$Z=4.677$	<0.001
Death, <i>n</i> (%)		0 (0.0)	0 (0.0)	1 (5.0)	–	1.000
Local complications, <i>n</i> (%)					–	0.371
Acute peripancreatic fluid accumulation			16 (80.0)	19 (95.0)		
Acute necrotic accumulation			2 (10.0)	2 (10.0)		
Pancreatic pseudocyst			2 (10.0)	7 (35.0)		
Walled-off necrosis			0 (0.0)	2 (10.0)		
Infected necrosis			0 (0.0)	1 (5.0)		

BMI: body mass index; D: one of CT degrees (A, B, C, D, and E); ACS: abdominal compartment syndrome; SIRS: systemic inflammatory response syndrome; ARDS: acute respiratory distress syndrome; AKI: acute kidney injury; CRP: C-reactive protein

^aRepresents $P<0.05$ vs NOR

^bRepresents $P<0.05$ vs MAP

^cRepresents $P<0.05$ vs MSAP

–Represents fisher test

(Supplementary Figure 1A), MSAP (Supplementary Figure 1B), and SAP (Supplementary Figure 1C). This finding indicated that heterogeneity between samples in AP was small.

Sample grouping analysis (PLS-DA analysis) showed a clear separation between the MAP/MSAP/SAP group and normal controls (Supplementary Figure 2). This suggested

that the structure of gut microbiota was significantly different between AP and NOR.

Gut Microbiota Species Annotation and Evaluation in MAP, MSAP, and SAP

Rank-abundance analysis showed that the gut microbiota species in MAP, MSAP, and SAP were abundant and evenly distributed (Supplementary Figure 3). As the number of samples increased, the total number of gut microbiota species increased, while the number of core gut microbiota species decreased in MAP, MSAP, and SAP in the pan/core analysis (Supplementary Figure 4). Diversity of community analysis (alpha diversity analysis) showed that community diversity was significantly increased ($0.01 < p < 0.05$) in Simpson index in MAP (Supplementary Figure 5A); community diversity and richness were significantly increased ($0.01 < p < 0.05$) in Shannon, Simpson, Ace, and Chao indexes in MSAP (Supplementary Figure 5B); community diversity was significantly increased ($0.01 < p < 0.05$) in Shannon and Simpson indexes in SAP (Supplementary Figure 5C).

Gut Microbiota Species Composition in MAP, MSAP, and SAP

The Venn diagram showed that there were 347, 923, and 663 unique gut microbiota species in MAP, MSAP, and SAP, respectively (Supplementary Figure 6). This finding showed an increase in gut microbiota species diversity in AP patients compared with normal control. The heat map of gut microbiota community composition richness information about the top 50 species in total abundance of OTU classification level in MAP, MSAP, and SAP is shown in Supplementary Figure 7A, 7B, and 7C, respectively. From the Circos diagram, we found that *Bacteroides* was dominant gut microbiota species in MAP at both genus and species levels (Fig. 1a); *Escherichis-Shigella* (a combination of *Escherichia* and *Shigella*) was dominant gut microbiota species in MSAP at both genus and species levels (Fig. 1b); in SAP, *Enterococcus* and *Escherichis-Shigella* were dominant gut microbiota species at genus and species levels, respectively (Fig. 1c).

Gut Microbiota Species Difference in MAP, MSAP, and SAP

In MAP, there were 101 and 213 significantly different gut microbiota species at the genus and species level, respectively. At the genus level, *Finegoldia* was the most significantly increased ($p \leq 0.001$) and *Eubacterium hallii* the most decreased ($p \leq 0.001$) species (Fig. 2a). At the species level, *Blautia* was the most significantly decreased ($p \leq 0.001$)

species (Fig. 2b). In MSAP, there were 113 and 217 significantly different gut microbiota species at the genus and species level, respectively. At the genus level, *Anaerococcus* was the most significantly increased ($p \leq 0.001$) and *Eubacterium hallii* the most decreased ($p \leq 0.001$) species (Fig. 3a). At the species level, *Finegoldia* was the most significantly increased ($p \leq 0.001$) species (Fig. 3b). In SAP, there were 113 and 208 significantly different gut microbiota species at the genus and species level, respectively. At the genus level, *Enterococcus* was the most significantly increased ($p \leq 0.001$) and *Eubacterium hallii* the most decreased ($p \leq 0.001$) species (Fig. 4a). At the species level, *Finegoldia* was the most significantly increased ($p \leq 0.001$) species (Fig. 4b). In addition, some skin bacteria (such as *Corynebacterium* and *Staphylococcus*) were also found in rectal swab samples of MAP (supplementary Table 1).

Potential Diagnostic Biomarkers for MAP, MSAP, and SAP

To identify potential diagnostic biomarkers for MAP, MSAP, and SAP, we conducted random forest feature selection procedures. All gut microbiota species were ranked according to the standardized drop in prediction accuracy. Three gut microbiota species, namely *Finegoldia*, *Eubacterium hallii*, and *Lachnospiraceae*, were considered as the optimal diagnostic biomarkers for MAP after subsequently adding one gut microbiota specie at a time in a top-down forward-wrapper approach (Fig. 5). *Eubacterium hallii* and *Anaerococcus* were considered as the optimal diagnostic biomarkers for MSAP after subsequently adding one gut microbiota specie at a time in a top-down forward-wrapper approach (Fig. 6) as many as 47 gut microbiota species (supplementary Table 2) were considered as the optimal diagnostic biomarkers for SAP after subsequently adding one gut microbiota specie at a time in a top-down forward-wrapper approach, and the top five gut microbiota species were *Eubacterium hallii*, *Ruminococcaceae*, *Lachnospiraceae_FCS020*, *Faecalibacterium*, and *Lachnospiraceae_NC2004* (Fig. 7).

Network Analysis in MAP, MSAP, and SAP

There were a total of two gut microbiota phyla, including *Firmicutes* and *Bacteroidetes* in the gut microbiota species correlation network map of MAP (Fig. 8a). Interestingly, there was a positive interaction between *Firmicutes* and *Bacteroidetes*. In addition, three gut microbiota phyla include *Firmicutes*, *Bacteroidetes*, and *Proteobacteria* in the gut microbiota species correlation network map of MSAP and SAP (Fig. 8b, c). However, there was no interaction between above mentioning three gut microbiota phyla.

A



B

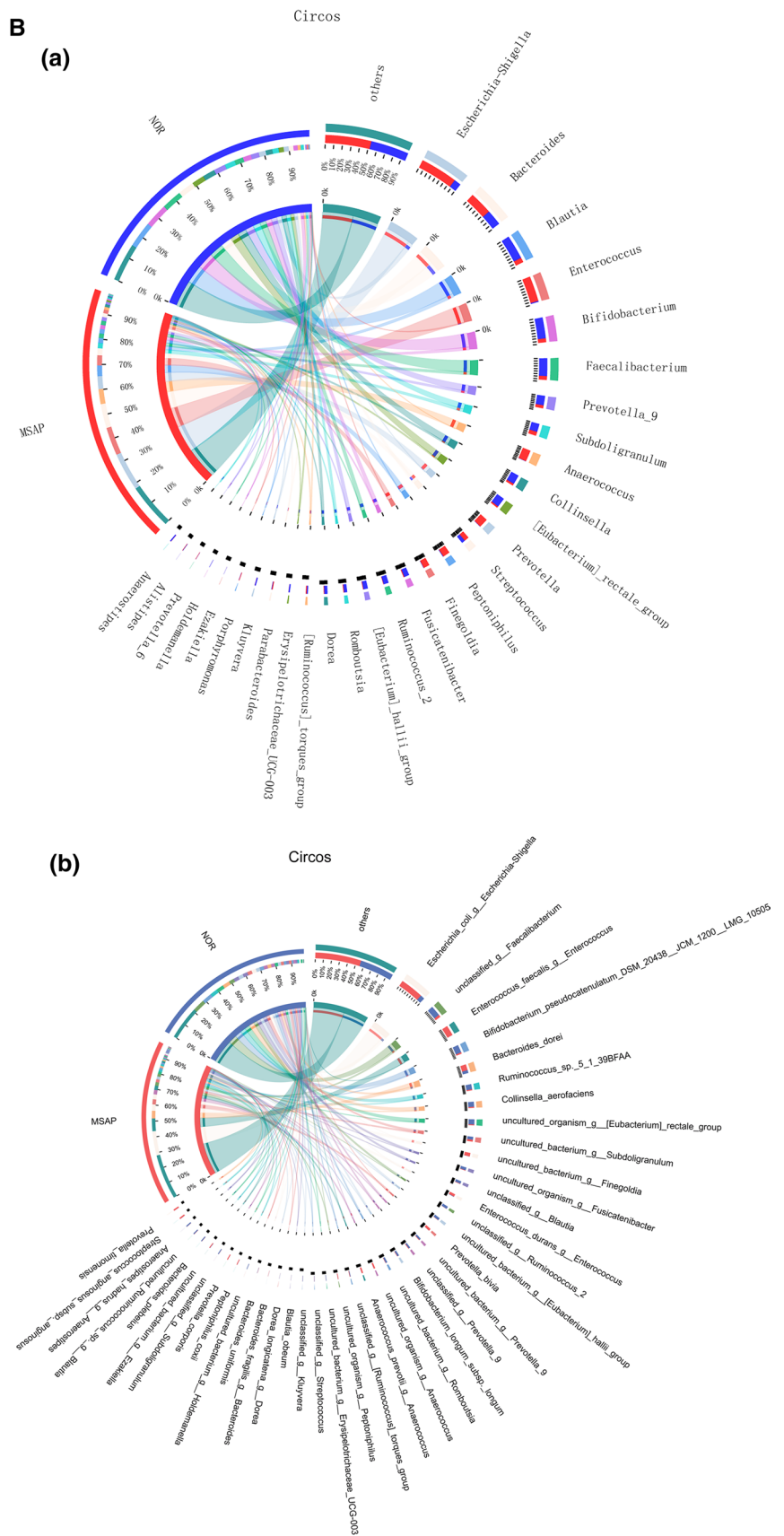
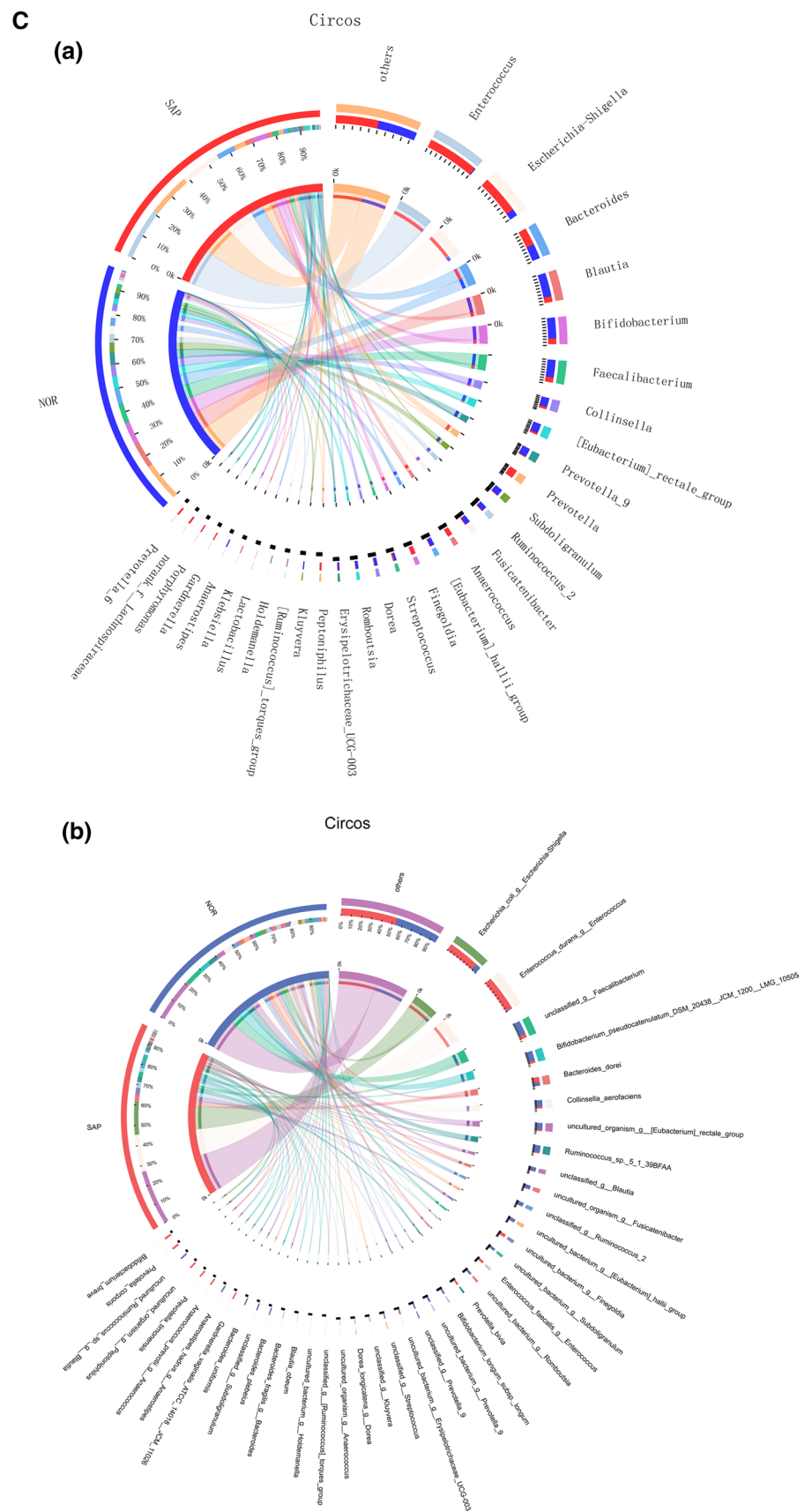


Fig. 1 (continued)



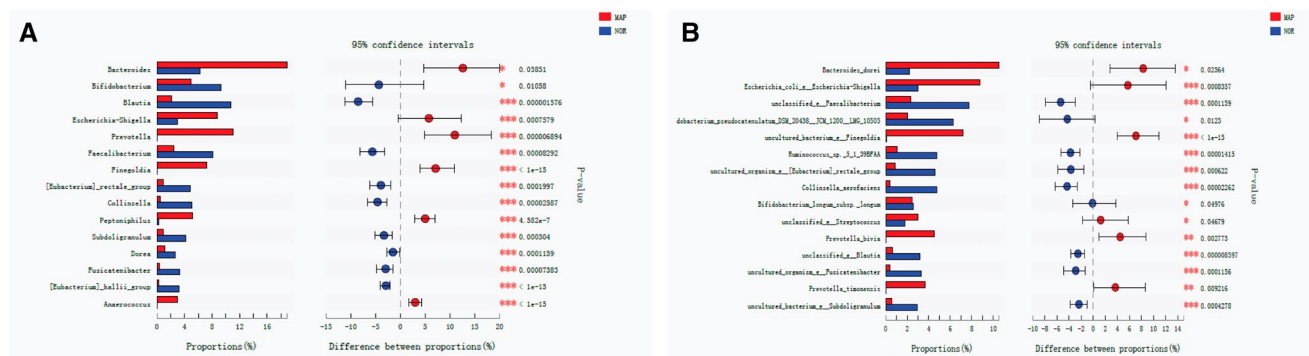


Fig. 2 Analysis of gut microbiota species difference in MAP at the genus and species levels in MAP. **a** At the genus level; **b** at the species level. On the left, X- and Y-axis represents the average relative abundance of the gut microbiota species in various groups and the

gut microbiota species names at a certain classification level, respectively. On the right, X- and Y-axis represents different gut microbiota species between various groups and p value of significance. $0.01 < p < 0.05$, $0.01 < p < 0.001$, $0.01 < p < 0.001$

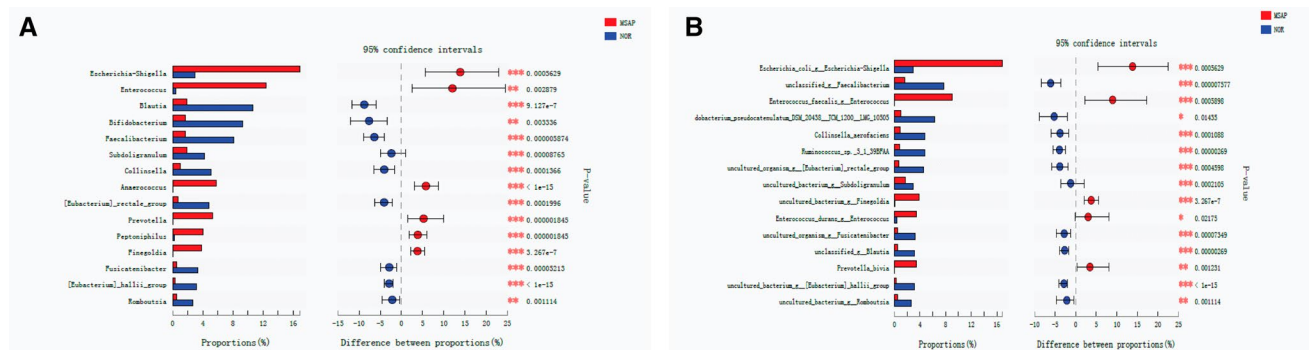


Fig. 3 Analysis of gut microbiota species difference in MAP at the genus and species levels in MSAP. **a** At the genus level; **b** at the species level. On the left, X- and Y-axis represents the average relative abundance of the gut microbiota species in various groups and the

gut microbiota species names at a certain classification level, respectively. On the right, X- and Y-axis represents different gut microbiota species between various groups and p value of significance. $0.01 < p < 0.05$, $0.01 < p < 0.001$, $0.01 < p < 0.001$

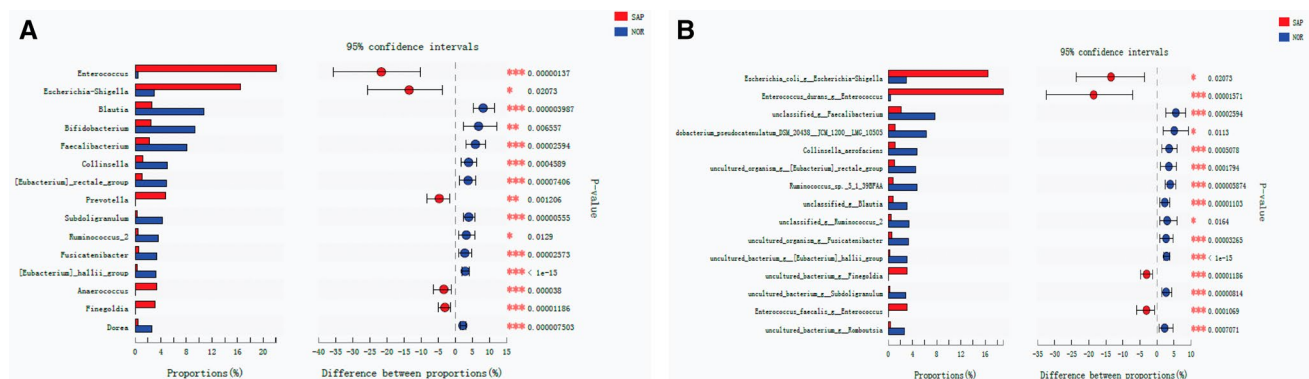


Fig. 4 Analysis of gut microbiota species difference in MAP at the genus and species levels in SAP. **a** At the genus level; **b** at the species level. On the left, X- and Y-axis represents the average relative abundance of the gut microbiota species in various groups and the

gut microbiota species names at a certain classification level, respectively. On the right, X- and Y-axis represents different gut microbiota species between various groups and p value of significance. $0.01 < p < 0.05$, $0.01 < p < 0.001$, $0.01 < p < 0.001$

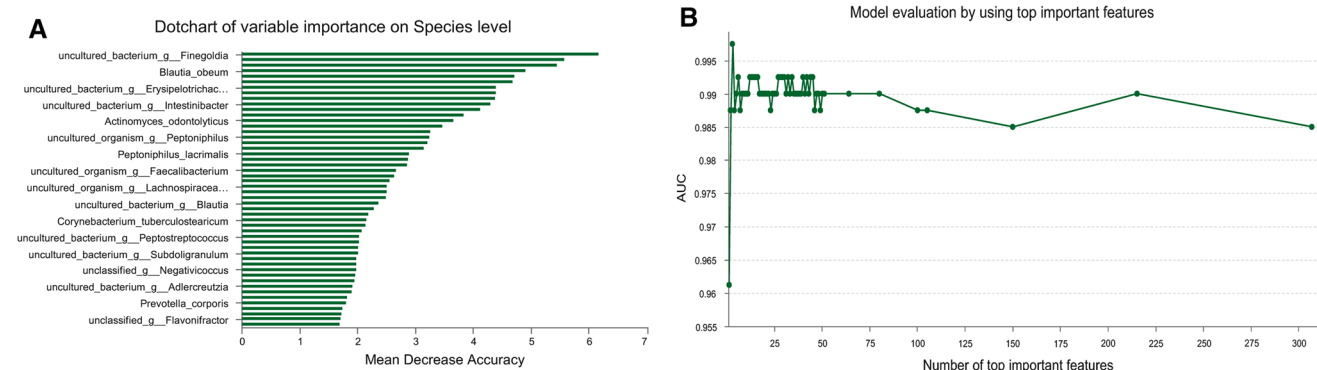


Fig. 5 Ranking of gut microbiota species and the tendency chart of AUC along with the increase in gut microbiota species in MAP. **a** The ranking of gut microbiota species; X- and Y-axis represents the measure of gut microbiota species importance and the gut microbiota species name in order of the importance, respectively. **b** The tendency chart of AUC along with the increase in gut microbiota spe-

cies. X- and Y-axis represents the number of gut microbiota species in the ranking of the importance and the AUC value of the random forest when the number of corresponding gut microbiota species is used, respectively. The solid point represents the point with the highest AUC value selected

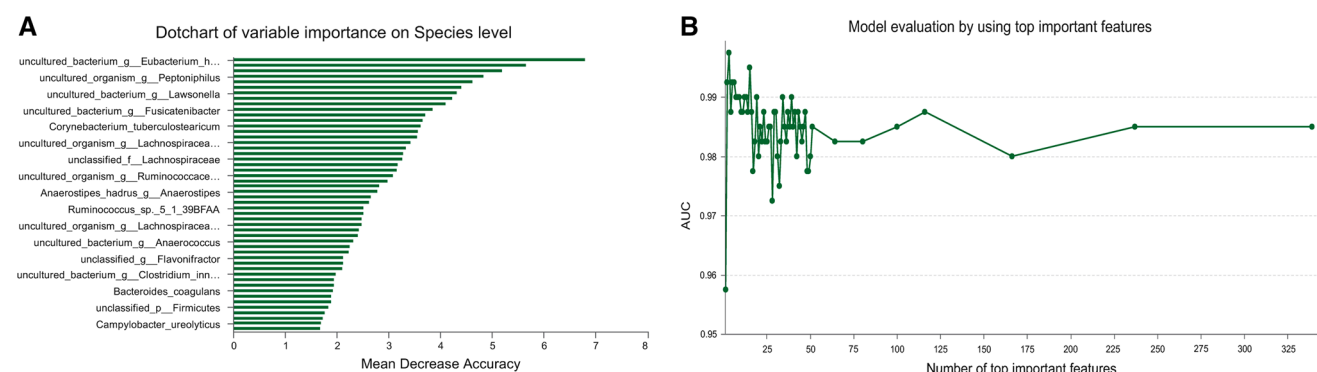


Fig. 6 Ranking of gut microbiota species and the tendency chart of AUC along with the increase in gut microbiota species in MAP. **a** The ranking of gut microbiota species; X- and Y-axis represents the measure of gut microbiota species importance and the gut microbiota species name in order of the importance, respectively. **b** The tendency chart of AUC along with the increase in gut microbiota spe-

cies. X- and Y-axis represents the number of gut microbiota species in the ranking of the importance and the AUC value of the random forest when the number of corresponding gut microbiota species is used, respectively. The solid point represents the point with the highest AUC value selected

Discussion

In this study, we performed the 16S rRNA sequencing approach to a cohort of 80 individuals including 60 AP patients (20 MAP, 20 MSAP, and 20 SAP) and 20 NOR, and we found significant dysbiosis of gut microbiota in AP patients. Moreover, the composition of gut microbiota differed significantly in the three grades of AP, implying an association between gut microbial dysbiosis and AP severity.

From the Circos diagram, we found that *Bacteroides*, *Escherichia-Shigella*, and *Enterococcus* were, respectively, the dominant gut microbial species in MAP, MSAP, and SAP, which is consistent with previous studies [10, 12, 22,

23]. The *Bacteroides*, an obligative anaerobic and gram-negative bacillus, plays a vital role to hydrolyze polysaccharides into short-chain fatty acids (SCFA) in the colon, which in turn inhibits anti-inflammatory factors and maintain gut barrier function [24]. *Bacteroides* has also been recognized as the most common microorganism isolated in infected pancreatic necrosis [25]. It is found that the proportions of *Bacteroides* are higher in patients with chronic pancreatitis [26], and treatment with antibiotics against *Bacteroides* is effective for infected pancreatic necrosis [22]. *Escherichia-Shigella*, an opportunistic pathogen, can disrupt the mucosal barrier and cause various adverse outcomes, such as hemorrhagic colitis and severe inflammation in the ileocolonic area [27, 28]. An increased abundance of *Escherichia-Shigella* has been found in the

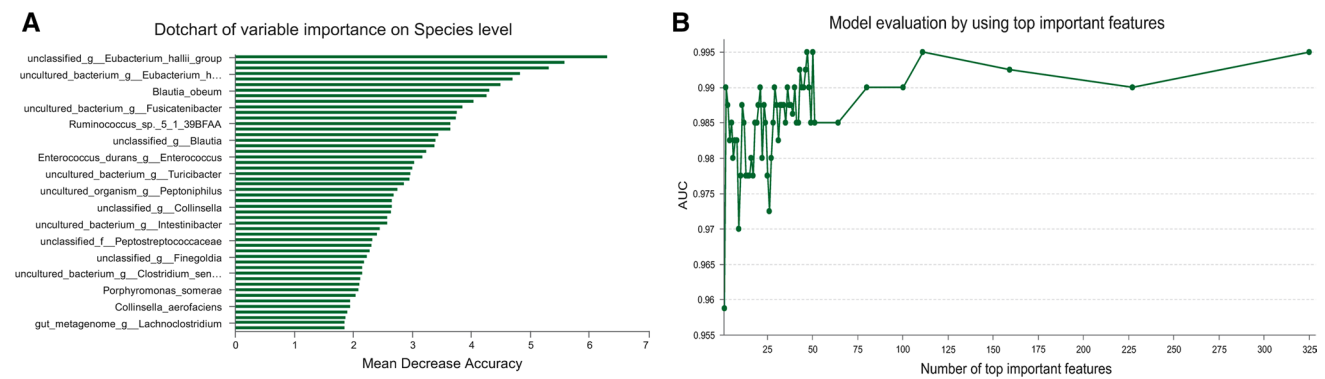


Fig. 7 Ranking of gut microbiota species and the tendency chart of AUC along with the increase in gut microbiota species in MAP. **a** The ranking of gut microbiota species; X- and Y-axis represents the measure of gut microbiota species importance and the gut microbiota species name in order of the importance, respectively. **b** The tendency chart of AUC along with the increase in gut microbiota spe-

cies. X- and Y-axis represents the number of gut microbiota species in the ranking of the importance and the AUC value of the random forest when the number of corresponding gut microbiota species is used, respectively. The solid point represents the point with the highest AUC value selected

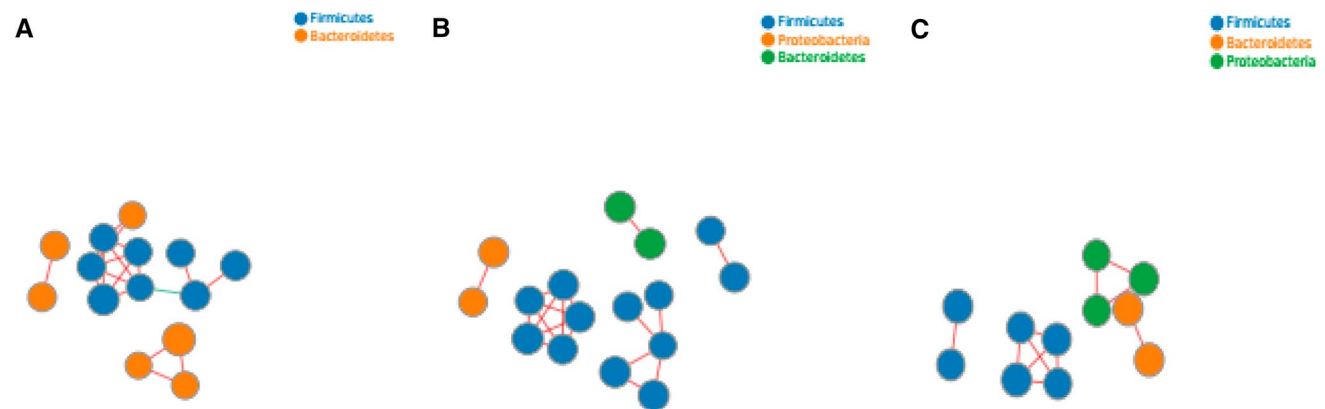


Fig. 8 Gut microbiota species correlation network analysis at the phylum level in MAP, MSAP, and SAP. The size of nodes indicates the abundance of gut microbiota species; the thickness of the line indicates the value of correlation coefficient; the color of the line

indicates the positive correlation, and the red and green indicate the positive and negative correlation between gut microbiota species, respectively

peripheral blood of patients with AP [29]. The relative abundance of *Enterococcus* (harmful pathogens) is related to infection and systemic inflammation [30]. The inoculation of *Il10*^{-/-} mice with *Enterococcus* species gives rise to characteristic chronic inflammatory diseases, supporting its pathogenic role [31]. It is worth mentioning that the population of intestinal *Enterococcus* is higher in the gut flora of SAP patients [32], which further demonstrated that *Enterococcus* might be a contributor to the severity of AP.

In the abundance analysis of gut microbial species, we found that *Finegoldia* and *Blautia* were, respectively, the most significantly increased ($p \leq 0.001$) and decreased ($p \leq 0.001$) gut microbiota species in MAP. *Finegoldia* inhabits the gastrointestinal as an opportunistic pathogen [33]. It is pointed out that *Finegoldia* was found in some inflammatory diseases, such as diverticulitis [34–39].

Blautia is the predominant taxa (harmless) in the intestinal tract of the healthy population [40]. Significant reduction in *Blautia* levels promotes the overgrowth of intestinal bacteria, increases intestinal permeability, and ultimately leads to higher concentrations of endotoxins and activation of inflammatory cascades [41–44]. This indicated that the disbalance of *Finegoldia* and *Blautia* might play a key role in the inflammatory process of MAP. In MSAP, *Anaerococcus* and *Eubacterium hallii* were, respectively, the most significantly increased ($p \leq 0.001$) and decreased ($p \leq 0.001$) gut microbiota species. The presence of *Anaerococcus* was also confirmed in the fluid of cystic pancreatic lesions, probably reflecting the indigenous microbiota of the pancreas [45]. In the human intestine, *Eubacterium hallii* is one of the most abundant butyrate (SCFA)-producing bacteria [46, 47]. Butyrate produces reuterin that regulates the homeostasis of

microbial metabolism, inhibits pathogens, and plays anti-inflammatory effects in the gut [48]. Our finding of increased *Anaerococcus* and decreased *Eubacterium hallii* suggested a perturbation in the mediation of the inflammatory response in patients with MSAP. Also, *Enterococcus* and *Eubacterium hallii* were, respectively, the most significantly increased ($p \leq 0.001$) and decreased ($p \leq 0.001$) gut microbiota species in SAP. Previous studies have demonstrated that *Enterococcus* could adhere and invade the surface of host cells, cross host epithelial barriers, and get access to internal organs, facilitating their dissemination in the host [10, 11]. SAP is characterized by local infection and sepsis, and *Enterococcus* has been identified in both pancreatic drainage and peripheral blood by culture or molecular techniques in such patients, indicating their translocation from the gut to the pancreas and systemic circulation as a contributor to sepsis in SAP patients [11, 23]. Thus, *Blautia*, *Anaerococcus*, and *Enterococcus* could serve as a gut microbial species abundance indicator of the MAP, MSAP, and SAP, respectively. We assume that the reduction in beneficial bacteria and the abundance of potentially harmful bacteria may promote bacterial and endotoxin translocation due to deficient gut barrier and thus contribute to the progression of AP.

In our study, the changed microbiota composition in different grades of AP is similar to previous studies [11, 23]. However, we employed techniques such as random forest model prediction and correlation network map to identify potentially useful biomarkers and explore relationships among gut microbiota species. Random forest is a machine learning algorithm, which can efficiently select the most appropriate biomarker for sample classification [49]. Furthermore, diagnostic ability of classification prediction was evaluated by the AUC value. To identify the optimal diagnostic biomarkers for MAP, MSAP, and SAP, the random forest feature selection procedure was performed. Three species, namely *Finegoldia*, *Eubacterium hallii*, and *Lachnospiraceae*, were revealed to be potential diagnostic biomarkers for MAP. *Lachnospiraceae*, beneficial gut microbial species, is a known butyrate producer and influences SCFA levels [50]. The protective function of *Lachnospiraceae* has been proven in inflammatory bowel disease and chronic alcoholic pancreatitis [51, 52]. In addition, two species of *Eubacterium hallii* and *Anaerococcus* were potentially useful biomarkers for MSAP.

In the correlation network map analysis, two phyla including *Bacteroidetes* and *Firmicutes* were identified in the gut microbiota species of MAP and *Proteobacteria* in both MSAP and SAP. Components of cultivable anaerobes from the *Bacteroidetes* are the primary inducers of pro-inflammatory process [53]. A previous study reported that the abundance of *Bacteroidetes* was decreased in patients with SAP [54]. *Firmicutes*, mainly consisted of gram-positive species of intestinal bacteria, produce SCFA [55,

56]. Diverse bacterial microbiota, dominated by the *Firmicutes*, was discovered in the peripheral blood of patients with SAP [57–59]. *Proteobacteria* was also detected to be more abundant in patients with chronic alcoholic pancreatitis and SAP [52, 60]. Interestingly, there was a positive interaction between *Bacteroidetes* and *Firmicutes*. This indicated that these two species might form a specific relationship, synergic with the destruction of the intestinal barrier in AP.

This is a pilot study with some limitations. Firstly, some skin bacteria (such as *Corynebacterium* and *Staphylococcus*) were found in our rectal swab samples (Supplementary Table 2). It is reported that a high relative abundance of skin bacteria, most frequently *Staphylococcus* and *Streptococcus*, is also observed in studies using stool samples [61]. In addition, skin bacteria *Corynebacterium* was more frequently found in rectal mucosal biopsies than in stool samples from healthy volunteers [62], suggesting that they may colonize in both anal skin and rectal mucosa. Therefore, further researches are eagerly awaited to address this issue in a larger number of comparative stool and rectal swab samples from AP patients. Secondly, although we identified some different gut microbiota in the progression of AP, it is not clear whether they are passive consequences or active contributors to severity. Further studies, such as the application of germ-free animals, are needed to investigate the potential mechanisms of gut dysbiosis in the worsening of AP. Lastly, our study was based on 16S rRNA sequencing, and a meta-genome approach should provide a more accurate profile of microbial composition and function in patients with AP. Such studies are eagerly awaited.

In summary, we have illustrated the disordered profiles of gut microbiota in different severity of AP, such as the increase in harmful pathogenic microbiota of *Enterococcus* in SAP and a decrease in beneficent microbiota of *Blautia* in MAP. Besides, potential diagnostic biomarkers for MAP (*Finegoldia*, *Eubacterium hallii*, and *Lachnospiraceae*) and MSAP (*Eubacterium hallii* and *Anaerococcus*) were further identified. The early depletion of SCFA-producing bacteria probably causes an impaired gut barrier and then drives the severity of AP. The overrepresentation of opportunistic pathogenic bacteria such as *Enterococcus* may be associated with pancreatic and systemic infection in SAP. The microbiota profile and its alterations may play an important role in the process of different severity of AP. Given all the limitations, our study could be regarded as a “concept research” with data laying the foundation for future descriptive and validation studies from a larger prospective multicenter project, which could help understand AP processes and severity and identify possible therapeutic targets.

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Conflict of interest The authors declare that they have no conflict of interest.

Research involving human participants All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This study was approved by the institutional review boards of Peking Union Medical College Hospital (No. JS 1826).

Informed consent Informed consent was obtained from all individual participants included in the study.

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